06_evaluation

Contents

1 06 Final Evaluation and Model Comparison **1**
1.1 1. Setup . . . 1
1.2 2. Load and Merge All Metrics . . . 2
1.3 3. LaTeX Comparison Tables . . . 3
1.4 4. Overfitting Analysis . . . 6
1.5 5. Best Model Predicted versus Actual . . . 8
1.6 6. Horizon Decay . . . 10
1.7 7. Naive Benchmark Comparison . . . 11
1.8 8. Consolidated Summary Table . . . 12
1.9 9. Sanity Checks . . . 13

1 06 Final Evaluation and Model Comparison

Authors: Aaron and Keira **Purpose:** Aggregate all model outputs from scripts 03, 04, and 05. Produce final comparison tables, overfitting diagnostics, horizon decay analysis, and naive benchmark evaluation. All plots follow QVRS (Quantitative Visualization and Reporting Standard) with Economist design conventions.

Depends on: - `output/predictions/ols_metrics.csv` (script 03) - `output/predictions/rf_xgboost_metrics.csv` (script 04) - `output/predictions/svm_ann_metrics.csv` (script 05)

Outputs: - Tables: `eval_rmse.tex`, `eval_mae.tex`, `eval_r2.tex`, `eval_da.tex`, `overfitting.tex`, `naive_comparison.tex`, `main_results.tex` - Figures: `overfitting_comparison.png`, `horizon_decay.png`, `best_model_pred_actual.png`

1.1 1. Setup

```r
library(tidyverse)
library(tidymodels)
library(here)
library(knitr)
library(kableExtra)

# Constants
TARGET_COLS  <- c("target_1d", "target_3d", "target_5d")
HORIZON_LABS <- c("1d",        "3d",        "5d")
MODEL_NAMES  <- c("OLS", "Random Forest", "XGBoost", "SVM", "ANN")

# QVRS colour tokens
CLR_RED      <- "#E3120B"  # Economist Red - benchmarks and emphasis
CLR_CHI_20   <- "#141F52"  # Chicago 20 - primary data
CLR_CHI_30   <- "#1F2E7A"  # Chicago 30 - secondary
CLR_CHI_45   <- "#2E45B8"  # Chicago 45 - tertiary
CLR_CHI_55   <- "#475ED1"  # Chicago 55 - quaternary
```

```r
CLR_LON_95   <- "#F2F2F2"  # London 95 - grid lines
CLR_LON_05   <- "#0D0D0D"  # London 5 - text
CLR_LON_20   <- "#333333"  # London 20 - subtitles
CLR_LON_35   <- "#595959"  # London 35 - captions
CLR_LA_95    <- "#F5F4EF"  # Los Angeles 95 - background
CLR_HK_45    <- "#1DC9A4"  # Hong Kong 45 - train bars
CLR_TOK_45   <- "#C91D42"  # Tokyo 45 - test bars

# QVRS ggplot2 theme - Economist standard
theme_quant <- function() {
  theme_minimal(base_size = 12) +
    theme(
      text = element_text(color = CLR_LON_05),
      plot.title = element_text(face = "bold", size = 14, margin = margin(b = 5)),
      plot.subtitle = element_text(size = 11, color = CLR_LON_20, margin = margin(b = 10)),
      plot.caption = element_text(size = 8, color = CLR_LON_35, hjust = 0, margin = margin(t = 10)),
      axis.title = element_text(size = 10, face = "bold"),
      axis.text = element_text(size = 9),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = CLR_LON_95, linewidth = 0.3),
      legend.position = "bottom",
      axis.ticks = element_blank(),
      plot.background = element_rect(fill = CLR_LA_95, color = NA),
      panel.background = element_rect(fill = CLR_LA_95, color = NA)
    )
}

# Ensure output directories exist
dir.create(here("paper", "tables"),  recursive = TRUE, showWarnings = FALSE)
dir.create(here("paper", "figures"),  recursive = TRUE, showWarnings = FALSE)
```

1.2 2. Load and Merge All Metrics

```r
ols_metrics   <- read_csv(here("output", "predictions", "ols_metrics.csv"),   show_col_types = FALSE)
rf_xgb_metrics <- read_csv(here("output", "predictions", "rf_xgboost_metrics.csv"), show_col_types = FA
svm_ann_metrics <- read_csv(here("output", "predictions", "svm_ann_metrics.csv"),   show_col_types = FA

# Separate naive rows from OLS metrics
naive_rows   <- ols_metrics %>%
  filter(Model == "Naive") %>%
  mutate(
    Horizon = case_when(
      is.numeric(Horizon) ~ paste0(Horizon, "d"),
      TRUE                ~ as.character(Horizon)
    )
  )
ols_only     <- ols_metrics %>% filter(Model == "OLS")

# Combine all model metrics (5 models x 3 horizons = 15 rows)
# Horizon may be numeric (1, 3, 5) or character ("1d", "3d", "5d") depending on
# how the source scripts wrote it. Normalise to character labels.
```

```r
all_metrics <- bind_rows(ols_only, rf_xgb_metrics, svm_ann_metrics) %>%
  mutate(
    Horizon = case_when(
      is.numeric(Horizon) ~ paste0(Horizon, "d"),
      TRUE                ~ as.character(Horizon)
    )
  ) %>%
  drop_na(Horizon, RMSE, MAE, R2, DA) %>%
  distinct(Model, Horizon, .keep_all = TRUE) %>%
  mutate(
    Model   = factor(Model, levels = MODEL_NAMES),
    Horizon = factor(Horizon, levels = HORIZON_LABS)
  )

cat("Total model-horizon rows:", nrow(all_metrics), "\n")
```

```
## Total model-horizon rows: 15
```

```r
cat("Naive benchmark rows    :", nrow(naive_rows), "\n")
```

```
## Naive benchmark rows    : 3
```

```r
cat("Models present:", paste(unique(all_metrics$Model), collapse = ", "), "\n")
```

```
## Models present: OLS, Random Forest, XGBoost, SVM, ANN
```

```r
cat("Horizons present:", paste(unique(all_metrics$Horizon), collapse = ", "), "\n")
```

```
## Horizons present: 1d, 3d, 5d
```

1.3 3. LaTeX Comparison Tables

Four tables. Each pivoted wide with models as rows and horizons as columns.

1.3.1 3A. RMSE Table

```r
rmse_wide <- all_metrics %>%
  mutate(Horizon = as.character(Horizon)) %>%
  select(Model, Horizon, RMSE) %>%
  pivot_wider(names_from = Horizon, values_from = RMSE, names_prefix = "h_", values_fn = first) %>%
  mutate(across(starts_with("h_"), ~round(.x, 5))) %>%
  rename(`1d` = h_1d, `3d` = h_3d, `5d` = h_5d)

latex_rmse <- kable(
  rmse_wide,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "1-Day", "3-Day", "5-Day")
)

rmse_note <- paste0(
  "Lower values indicate better predictive accuracy. ",
```

```
  "All models estimated on 80/20 chronological split. ",
  "Training period ends 2024-10-21. Test period: 2024-10-22 to 2026-04-01."
)

latex_rmse <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Out-of-Sample RMSE by Model and Prediction Horizon}\n",
  "\\label{tab:eval_rmse}\n",
  as.character(latex_rmse), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", rmse_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_rmse, here("paper", "tables", "eval_rmse.tex"))
message("Saved: paper/tables/eval_rmse.tex")
```

1.3.2 3B. MAE Table

```
mae_wide <- all_metrics %>%
  mutate(Horizon = as.character(Horizon)) %>%
  select(Model, Horizon, MAE) %>%
  pivot_wider(names_from = Horizon, values_from = MAE, names_prefix = "h_", values_fn = first) %>%
  mutate(across(starts_with("h_"), ~round(.x, 5))) %>%
  rename(`1d` = h_1d, `3d` = h_3d, `5d` = h_5d)

latex_mae <- kable(
  mae_wide,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "1-Day", "3-Day", "5-Day")
)

mae_note <- paste0(
  "Mean Absolute Error. Lower values indicate better predictive accuracy. ",
  "MAE is less sensitive to outlier predictions than RMSE."
)

latex_mae <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Out-of-Sample MAE by Model and Prediction Horizon}\n",
  "\\label{tab:eval_mae}\n",
  as.character(latex_mae), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", mae_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
```

```
  "\\end{table}"
)

writeLines(latex_mae, here("paper", "tables", "eval_mae.tex"))
message("Saved: paper/tables/eval_mae.tex")
```

1.3.3 3C. R-squared Table

```
r2_wide <- all_metrics %>%
  mutate(Horizon = as.character(Horizon)) %>%
  select(Model, Horizon, R2) %>%
  pivot_wider(names_from = Horizon, values_from = R2, names_prefix = "h_", values_fn = first) %>%
  mutate(across(starts_with("h_"), ~round(.x, 5))) %>%
  rename(`1d` = h_1d, `3d` = h_3d, `5d` = h_5d)

latex_r2 <- kable(
  r2_wide,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "1-Day", "3-Day", "5-Day")
)

r2_note <- paste0(
  "Out-of-sample R-squared computed as 1 minus the ratio of model MSE to mean MSE. ",
  "Negative values indicate the model performs worse than predicting the mean return."
)

latex_r2 <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Out-of-Sample R-squared by Model and Prediction Horizon}\n",
  "\\label{tab:eval_r2}\n",
  as.character(latex_r2), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", r2_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_r2, here("paper", "tables", "eval_r2.tex"))
message("Saved: paper/tables/eval_r2.tex")
```

1.3.4 3D. Directional Accuracy Table

```
da_wide <- all_metrics %>%
  mutate(Horizon = as.character(Horizon)) %>%
  select(Model, Horizon, DA, Naive_DA) %>%
  pivot_wider(
    names_from  = Horizon,
```

```r
    values_from = c(DA, Naive_DA),
    names_sep   = "_",
    values_fn   = first
  ) %>%
  mutate(across(where(is.double), ~round(.x, 4)))

# Reorder columns: Model, DA_1d, DA_3d, DA_5d, Naive_1d, Naive_3d, Naive_5d
da_wide <- da_wide %>%
  select(Model, DA_1d, DA_3d, DA_5d, Naive_DA_1d, Naive_DA_3d, Naive_DA_5d)

latex_da <- kable(
  da_wide,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "DA 1d", "DA 3d", "DA 5d", "Naive 1d", "Naive 3d", "Naive 5d")
)

da_note <- paste0(
  "DA = fraction of test-set observations where predicted sign matches realised sign. ",
  "Naive benchmark uses the previous day BPI log return (lag1). ",
  "Values above 0.50 indicate better-than-random directional prediction."
)

latex_da <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Directional Accuracy by Model and Prediction Horizon}\n",
  "\\label{tab:eval_da}\n",
  as.character(latex_da), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", da_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_da, here("paper", "tables", "eval_da.tex"))
message("Saved: paper/tables/eval_da.tex")
```

1.4 4. Overfitting Analysis

```r
overfit_df <- all_metrics %>%
  mutate(
    RMSE_Gap = round(Train_RMSE - RMSE, 5),
    Model    = as.character(Model)
  )

# Bar plot: Train RMSE vs Test RMSE by model, faceted by horizon
overfit_plot_data <- all_metrics %>%
```

```r
  select(Model, Horizon, RMSE, Train_RMSE) %>%
  pivot_longer(cols = c(RMSE, Train_RMSE), names_to = "Set", values_to = "Value") %>%
  mutate(
    Set = if_else(Set == "Train_RMSE", "In-Sample", "Out-of-Sample"),
    Set = factor(Set, levels = c("In-Sample", "Out-of-Sample")),
    Model = factor(Model, levels = MODEL_NAMES)
  )

# Model colour mapping
model_colors <- setNames(
  c(CLR_CHI_20, CLR_CHI_30, CLR_CHI_45, CLR_CHI_55, CLR_RED),
  MODEL_NAMES
)

overfit_plot <- ggplot(overfit_plot_data, aes(x = Model, y = Value, fill = Set)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  facet_wrap(~Horizon, nrow = 1, labeller = labeller(Horizon = c(
    "1d" = "1-Day Horizon", "3d" = "3-Day Horizon", "5d" = "5-Day Horizon"
  ))) +
  scale_fill_manual(values = c("In-Sample" = CLR_HK_45, "Out-of-Sample" = CLR_TOK_45)) +
  labs(
    title    = "Train and Test RMSE Divergence Signals Overfitting",
    subtitle = "In-sample error is systematically lower than out-of-sample error across all models",
    x        = NULL,
    y        = "RMSE (Daily Log Return)",
    fill     = NULL,
    caption  = "Source. Authors calculations. Training period ends 2024-10-21. Test period 2024-10-22 t
  ) +
  theme_quant() +
  theme(
    strip.text = element_text(face = "bold", size = 10),
    legend.position = "bottom",
    axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1, size = 8)
  )

ggsave(
  here("paper", "figures", "overfitting_comparison.png"),
  overfit_plot,
  width = 11, height = 5, dpi = 300
)
message("Saved: paper/figures/overfitting_comparison.png")

# LaTeX table
overfit_tex <- overfit_df %>%
  select(Model, Horizon, Train_RMSE, RMSE, RMSE_Gap) %>%
  mutate(
    across(c(Train_RMSE, RMSE, RMSE_Gap), ~round(.x, 5))
  )

latex_overfit <- kable(
  overfit_tex,
  format    = "latex",
  booktabs  = TRUE,
```

```r
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "Horizon", "Train RMSE", "Test RMSE", "Gap")
)

overfit_note <- paste0(
  "Gap = Train RMSE minus Test RMSE. Positive values indicate the model fits ",
  "training data more closely than it generalises to unseen observations. ",
  "Large gaps relative to test RMSE suggest overfitting."
)

latex_overfit <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Overfitting Diagnostic: Train versus Test RMSE}\n",
  "\\label{tab:overfitting}\n",
  as.character(latex_overfit), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", overfit_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_overfit, here("paper", "tables", "overfitting.tex"))
message("Saved: paper/tables/overfitting.tex")
```

1.5 5. Best Model Predicted versus Actual

```r
# Identify best model by lowest test RMSE on the 1-day horizon
best_model_name <- all_metrics %>%
  filter(Horizon == "1d") %>%
  slice_min(RMSE, n = 1) %>%
  pull(Model) %>%
  as.character()

cat("Best model (1-day RMSE):", best_model_name, "\n")
```

```
## Best model (1-day RMSE): ANN
```

```r
# Map model name to saved model file prefix
model_file_prefix <- case_when(
  best_model_name == "OLS"           ~ "ols",
  best_model_name == "Random Forest" ~ "rf",
  best_model_name == "XGBoost"       ~ "xgboost",
  best_model_name == "SVM"           ~ "svm",
  best_model_name == "ANN"           ~ "ann",
  TRUE                               ~ "unknown"
)

# Load the best model and generate test predictions
best_model_path <- here("output", "models", paste0(model_file_prefix, "_1d.rds"))
```

```r
if (file.exists(best_model_path)) {
  best_fit <- readRDS(best_model_path)

  # Load features for test set reconstruction
  features <- read_csv(here("data", "processed", "features.csv"), show_col_types = FALSE)
  n_total  <- nrow(features)
  n_train  <- floor(0.80 * n_total)
  test_df  <- features[(n_train + 1L):n_total, ]

  # Determine predictor columns based on model type
  if (best_model_name == "OLS") {
    pred_cols <- c(
      "r_bpi", "r_bdo", "r_psei", "r_fx", "r_us10y",
      "ema12", "ema26", "ma_ratio",
      "lag1", "lag2", "lag3", "mom5", "mom10", "rsi14", "macd", "macd_signal",
      "vol5", "vol10", "vol_ratio",
      "lag_psei", "interact", "vol5_sq", "hi_vol"
    )
  } else {
    pred_cols <- c(
      "r_bpi", "r_bdo", "r_psei", "r_fx", "r_us10y",
      "ma5", "ma10", "ma20", "ema12", "ema26", "ma_ratio",
      "lag1", "lag2", "lag3", "mom5", "mom10", "rsi14", "macd", "macd_signal",
      "vol5", "vol10", "vol20", "vol_ratio",
      "lag_psei", "interact", "vol5_sq", "hi_vol"
    )
  }

  test_h <- test_df %>% select(all_of(c("target_1d", pred_cols)))

  if (best_model_name == "OLS") {
    test_pred <- predict(best_fit, newdata = test_h)
  } else {
    test_pred <- predict(best_fit, new_data = test_h)$.pred
  }

  best_actual <- test_h$target_1d
} else {
  # Fallback: use predictions from CSV
  cat("Model file not found. Falling back to CSV predictions.\n")
  pred_csv <- read_csv(here("output", "predictions", "ols_predictions.csv"), show_col_types = FALSE)
  best_actual <- pred_csv$actual_target_1d
  test_pred   <- pred_csv[[paste0("ols_pred_1d")]]
}

# Scatter plot
best_plot <- ggplot(
  tibble(Actual = best_actual, Predicted = as.numeric(test_pred)),
  aes(x = Actual, y = Predicted)
) +
  geom_point(color = CLR_CHI_20, alpha = 0.5, size = 1.2) +
  geom_abline(intercept = 0, slope = 1, color = CLR_RED, linetype = "dashed", linewidth = 0.8) +
  labs(
```

```
    title    = paste(best_model_name, "Predictions Cluster Around Zero"),
    subtitle = paste0("1-Day Horizon. Test RMSE: ",
                      round(all_metrics %>% filter(Model == best_model_name, Horizon == "1d") %>% pull(
    x        = "Actual Log Return",
    y        = "Predicted Log Return",
    caption  = paste0("Source. Authors calculations. ", best_model_name,
                      " model trained on 80 percent sample. Test period 2024-10-22 to 2026-04-01.")
  ) +
  theme_quant()

ggsave(
  here("paper", "figures", "best_model_pred_actual.png"),
  best_plot,
  width = 8, height = 6, dpi = 300
)
message("Saved: paper/figures/best_model_pred_actual.png")
```

1.6 6. Horizon Decay

```
# Line plot: RMSE by horizon for each model, with naive as benchmark
horizon_plot_data <- all_metrics %>%
  mutate(Horizon_num = as.numeric(factor(Horizon, levels = HORIZON_LABS)))

# Add naive benchmark
naive_plot_data <- naive_rows %>%
  drop_na(Horizon, RMSE) %>%
  mutate(
    Horizon_num = as.numeric(factor(Horizon, levels = HORIZON_LABS)),
    Model = "Naive Benchmark"
  ) %>%
  select(Model, Horizon_num, RMSE)

horizon_plot <- ggplot() +
  geom_line(
    data = horizon_plot_data,
    aes(x = Horizon_num, y = RMSE, color = Model, group = Model),
    linewidth = 0.8
  ) +
  geom_point(
    data = horizon_plot_data,
    aes(x = Horizon_num, y = RMSE, color = Model),
    size = 2
  ) +
  geom_line(
    data = naive_plot_data,
    aes(x = Horizon_num, y = RMSE),
    color = CLR_RED,
    linewidth = 0.8,
    linetype = "dashed"
  ) +
  geom_point(
```

```r
    data = naive_plot_data,
    aes(x = Horizon_num, y = RMSE),
    color = CLR_RED,
    size = 2
  ) +
  scale_x_continuous(breaks = 1:3, labels = c("1-Day", "3-Day", "5-Day")) +
  scale_color_manual(values = model_colors) +
  labs(
    title    = "Predictive Accuracy Decays as the Forecast Horizon Extends",
    subtitle = "RMSE rises from 1-day to 5-day for all models. Naive benchmark shown for reference.",
    x        = "Forecast Horizon",
    y        = "RMSE (Daily Log Return)",
    color    = NULL,
    caption  = "Source. Authors calculations. Dashed red line is the naive benchmark using lagged retur
  ) +
  theme_quant() +
  theme(legend.position = "bottom")

ggsave(
  here("paper", "figures", "horizon_decay.png"),
  horizon_plot,
  width = 8, height = 6, dpi = 300
)
message("Saved: paper/figures/horizon_decay.png")
```

1.7 7. Naive Benchmark Comparison

```r
# Merge model DA with naive DA
naive_comp <- all_metrics %>%
  select(Model, Horizon, DA, Naive_DA) %>%
  mutate(
    Beats_Naive = if_else(DA > Naive_DA, "Yes", "No"),
    DA_Diff     = round(DA - Naive_DA, 4)
  ) %>%
  arrange(Model, Horizon)

latex_naive <- kable(
  naive_comp,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "Horizon", "DA", "Naive DA", "Beats Naive", "DA Difference")
)

naive_note <- paste0(
  "DA Difference = Model DA minus Naive DA. Positive values indicate the model ",
  "outperforms the naive benchmark. The naive benchmark uses the previous day ",
  "BPI log return (lag1) as the prediction."
)
```

```r
latex_naive <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Directional Accuracy versus Naive Benchmark}\n",
  "\\label{tab:naive_comparison}\n",
  as.character(latex_naive), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", naive_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_naive, here("paper", "tables", "naive_comparison.tex"))
message("Saved: paper/tables/naive_comparison.tex")
```

1.8 8. Consolidated Summary Table

Compact master table with RMSE and DA for quick reference in the paper.

```r
main_tbl <- all_metrics %>%
  mutate(Horizon = as.character(Horizon)) %>%
  select(Model, Horizon, RMSE, DA) %>%
  mutate(
    RMSE = round(RMSE, 5),
    DA   = round(DA, 4)
  ) %>%
  pivot_wider(
    names_from  = Horizon,
    values_from = c(RMSE, DA),
    names_sep   = "_",
    values_fn   = first
  ) %>%
  select(
    Model,
    RMSE_1d, DA_1d,
    RMSE_3d, DA_3d,
    RMSE_5d, DA_5d
  )

latex_main <- kable(
  main_tbl,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "RMSE 1d", "DA 1d", "RMSE 3d", "DA 3d", "RMSE 5d", "DA 5d")
)

main_note <- paste0(
  "RMSE = Root Mean Squared Error. DA = Directional Accuracy. ",
  "Negative R-squared values indicate the model performs worse than predicting the mean return."
```

```r
)

latex_main <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\begin{threeparttable}\n",
  "\\caption{Summary of Out-of-Sample RMSE and Directional Accuracy}\n",
  "\\label{tab:main_results}\n",
  as.character(latex_main), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", main_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{threeparttable}\n",
  "\\end{table}"
)

writeLines(latex_main, here("paper", "tables", "main_results.tex"))
message("Saved: paper/tables/main_results.tex")
```

1.9 9. Sanity Checks

```r
cat("=== SANITY CHECKS ===\n\n")
```

```
## === SANITY CHECKS ===
```

```r
# 1. Row count: 5 models x 3 horizons = 15
expected_rows <- 15
actual_rows   <- nrow(all_metrics)
cat("1. Model-horizon rows:", actual_rows,
    if (actual_rows == expected_rows) "(PASS)\n"
    else paste0("(FAIL - expected", expected_rows, ", got", actual_rows, ")\n"))
```

```
## 1. Model-horizon rows: 15 (PASS)
```

```r
# 2. All 5 models present
models_present <- sort(unique(as.character(all_metrics$Model)))
models_expected <- sort(MODEL_NAMES)
cat("2. Models present:", paste(models_present, collapse = ", "), "\n")
```

```
## 2. Models present: ANN, OLS, Random Forest, SVM, XGBoost
```

```r
cat("   Models expected:", paste(models_expected, collapse = ", "), "\n")
```

```
##    Models expected: ANN, OLS, Random Forest, SVM, XGBoost
```

```r
cat("   ", if (identical(models_present, models_expected)) "PASS\n" else "FAIL\n")
```

```
##     PASS
```

```r
# 3. All 3 horizons present
horizons_present <- sort(unique(as.character(all_metrics$Horizon)))
horizons_expected <- sort(HORIZON_LABS)
cat("3. Horizons present:", paste(horizons_present, collapse = ", "),
    if (identical(horizons_present, horizons_expected)) "(PASS)\n"
    else "(FAIL)\n")
```

```
## 3. Horizons present: 1d, 3d, 5d (PASS)
```

```r
# 4. No NA in key metrics
na_check <- all_metrics %>%
  summarise(
    na_RMSE = sum(is.na(RMSE)),
    na_MAE  = sum(is.na(MAE)),
    na_R2   = sum(is.na(R2)),
    na_DA   = sum(is.na(DA))
  )
if (all(na_check == 0)) {
  cat("4. No NA in metrics: PASS\n")
} else {
  cat("4. NA in metrics: FAIL\n")
  print(na_check)
}
```

```
## 4. No NA in metrics: PASS
```

```r
# 5. Output files exist
output_files <- c(
  # Tables
  here("paper", "tables", "eval_rmse.tex"),
  here("paper", "tables", "eval_mae.tex"),
  here("paper", "tables", "eval_r2.tex"),
  here("paper", "tables", "eval_da.tex"),
  here("paper", "tables", "overfitting.tex"),
  here("paper", "tables", "naive_comparison.tex"),
  here("paper", "tables", "main_results.tex"),
  # Figures
  here("paper", "figures", "overfitting_comparison.png"),
  here("paper", "figures", "horizon_decay.png"),
  here("paper", "figures", "best_model_pred_actual.png")
)

cat("5. Output files:\n")
```

```
## 5. Output files:
```

```r
for (f in output_files) {
  exists_flag <- file.exists(f)
  cat("   ", basename(f), ": ", if (exists_flag) "PASS\n" else "FAIL (missing)\n")
}
```

```
##     eval_rmse.tex :  PASS
##     eval_mae.tex :  PASS
##     eval_r2.tex :  PASS
##     eval_da.tex :  PASS
##     overfitting.tex :  PASS
##     naive_comparison.tex :  PASS
##     main_results.tex :  PASS
##     overfitting_comparison.png :  PASS
##     horizon_decay.png :  PASS
##     best_model_pred_actual.png :  PASS
```

```r
cat("\n=== All checks complete ===\n")
```

```
##
```

```
## === All checks complete ===
```